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Distributional implications of selection into heating-system retrofit subsidies

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ABSTRACT

As the shift away from fossil fuel-based heating systems has become a political priority in many countries, subsidies for energy renovation measures have emerged as popular policy tools to support an equitable energy transition. In this paper, we use high-quality administrative data to study the distributional implications of self-selection into three heating-system retrofit subsidies aimed at phasing out oil heating in Finnish residential buildings during 2020–2022. By comparing subsidy applicants with households that carried out similar renovations before the subsidies were introduced and with those that have yet to renovate, we show that subsidy uptake is concentrated among better-off households — mirroring the pattern observed for earlier, unsubsidised renovations. Consequently, the studied subsidy programmes have not substantially altered the types of households replacing oil heating. We subsequently compare the applicants across the three distinct subsidy programmes and find that a tax-credit appears a particularly regressive form of subsidy while a simpler lump-sum instrument attracts a broader range of applicants. Our setting provides a valuable opportunity to isolate how subsidy design shapes distributional outcomes, offering insights for the design of future energy renovation policies.

1. Introduction

Residential heating is a major driver of energy use and emissions worldwide, with almost half of buildings' energy demand in 2022 used for space and water heating, producing over 4.1 Gt CO_2 [1]. Worldwide, around 40% of households require space heating during part of the year, making it a dominant component of home energy use, especially in colder climates [1]. For example, in Finland, building heating accounted for approximately 27% of total energy consumption in 2022 [2]. With most existing buildings expected to remain in use through 2050, reducing energy consumption and emissions of the current building stock is imperative [3].

Energy renovations, particularly heating-system retrofits[1] that replace fossil-fuelled systems with more sustainable alternatives, are a way to achieve significant energy and emissions savings. To accelerate the transition, energy renovation subsidies that support households in covering renovation costs have become important policy instruments. For instance, public grant programmes are now utilised in almost all EU Member States [6]. As these programmes have expanded, their justice and vulnerability implications have become an increasingly important issue. Vulnerability arises when the benefits and burdens of energy systems and transitions are distributed unevenly or when some groups face barriers to participating in and benefiting from energy-system change [7,8]. While decarbonisation can reduce the socio-economic and environmental risks associated with continued fossil-fuel use, low-carbon transitions and related policies can also create new vulnerabilities and injustices [8]. In residential heating, these may emerge when financial, informational, or technological barriers prevent households from replacing fossil-fuel heating with cleaner, lower-cost alternatives, excluding them from the benefits of the energy transition [9].

Consistent with these concerns, existing research has highlighted that subsidies directed at household energy efficiency often fail to reach an efficient, let alone just outcome. For example, Borenstein and Davis [10] find that in a U.S. tax credit programme for clean energy investments – including energy efficiency measures, solar panels, and electric vehicles – the bottom three income quintiles received roughly 10% of all credits, while the top quintile captured approximately 60%. Similar distributional patterns have been documented in a range of contexts [e.g., 11–15]. This type of evidence, however, comes predominantly from studies of a single subsidy type, making it

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[1] The terms "renovation" and "retrofit" are used almost interchangeably in the literature, although retrofits typically refer to a narrower set of measures focused on replacing or upgrading existing systems [4,5]. We use energy renovation as the umbrella term that includes retrofit measures.

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difficult to assess whether distributional outcomes reflect a particular instrument or a broader pattern. Evidence from tax-credit schemes, for instance, suggests that lower-income households and those with limited tax liability are structurally less able to benefit [16], pointing to eligibility constraints — but whether similar dynamics operate under other instrument types remains unclear. Moreover, inequality in uptake has been examined predominantly through the lens of income, while variation along other dimensions and the mechanisms behind unequal participation remain understudied.

In this paper we examine the distributional implications of subsidies for heating-system retrofits, focusing on self-selection – the feature that households themselves choose whether to apply – as a mechanism. We study this using high-quality administrative data for a wave of three subsidy programmes – a tax credit, a lump-sum transfer, and a renovation cost-share subsidy – implemented in Finland in 2020–2022.

All three instruments subsidised the replacement of oil heating with more sustainable alternatives. The focus on oil heating replacements is justified for multiple reasons. First, the fact that three parallel policy instruments simultaneously targeted the same renovation action provides a rare opportunity to compare different subsidy designs. Second, we have both high-quality data on the renovation action and the ability to identify the target population for whom it is relevant. Third, in the Finnish context, oil-boiler replacement has been identified as a highly cost-effective measure for reducing emissions [17].

Our analysis focuses on the distribution of subsidy uptake among owners of detached and semi-detached houses, the segment where oil heating is most concentrated and which accounted for the vast majority of applications. We address two research questions using discrete choice models: (i) How do the socioeconomic characteristics of applicants to heating-system retrofit subsidies differ from those of remaining oil heaters and earlier renovators? (ii) How do the applicants differ between the three subsidy programmes? We address these questions in two steps: first comparing subsidy applicants to oil-heating households that did not apply for any subsidy, and then comparing applicants across the three subsidy programmes.

We find that, compared to the average homeowner, oil-heating households exhibit characteristics commonly associated with energy-transition-related vulnerabilities, as they tend to be older, have lower income and education levels, and are more likely to reside in rural areas [8,18]. Because oil heating is concentrated among these more vulnerable groups, a subsidy targeted at oil-heating households could ostensibly reduce inequalities in sustainable heating system adoption. However, we uncover evidence that this level of targeting alone is insufficient to achieve a non-regressive outcome. We find that subsidies are predominantly taken up by higher-income households, who also tend to be younger, more urban and more highly educated than other oil heaters. They are more likely to live in newer houses, in which they have been living for a shorter period of time, and less likely to be renting out their house. By comparing the three subsidies, we demonstrate that subsidy design shapes distributional outcomes. In our setting we find that a tax credit system is a particularly regressive form of energy renovation subsidy, while a simpler lump-sum subsidy attracted more applicants from lower income and education brackets, suggesting that it could lead to a more equitable outcome.

This paper makes three contributions to the literature on energy renovation subsidies. First, our individual-level administrative data make it possible to move beyond income as the sole dimension of inequality, shedding light on how characteristics such as age, tenure, and housing condition shape who takes up subsidies — aspects that have received little systematic attention. Second, because the data cover not only applicants but also households that chose not to apply and those that had already renovated without subsidies, we can distinguish self-selection into subsidies from selection into renovation more broadly. This allows us to examine whether subsidies draw in a different type of household than those who would have renovated anyway, and to focus on mechanisms that operate specifically through the subsidy rather than broader renovation incentives.

Third, we examine a unique setting of three concurrently implemented programmes with distinct designs, all targeting the same population. Because the programmes operated in the same context and time period, differences in outcomes can be more credibly attributed to design rather than to the broader environment. We are thus able to compare programmes and isolate the relationship between subsidy design and distributional outcomes. This speaks to whether the regressive patterns documented in the literature are a feature of particular instrument designs or a characteristic of renovation subsidies more broadly. These findings offer particularly important lessons now as the new EU Emissions Trading System 2 (ETS2) will price CO_2 emissions from residential heating, placing increasing pressure on households using fossil fuel-based heating. At the same time, substantial funds from ETS2 should be directed toward vulnerable households, raising the question of how subsidies should be designed to support affected households.

The remainder of this paper is structured as follows: Section 2 reviews relevant previous research. Section 3 presents the institutional background of residential heating in Finland and the three subsidy programmes. Section 4 describes the data and methods used in the analysis. Section 5 reports the results where we first examine the socio-economic profile of households targeted by the subsidies and the benefit distribution, then analyse the determinants of applying overall as well as differences across the three programmes, and finally explore the potential reasons behind these patterns. Section 6 discusses the results and reflects on their implications. Section 7 concludes and states the policy implications of our findings

2. Previous research

Energy renovations offer private economic benefits by reducing energy costs and increasing property values, as well as social and environmental benefits, such as improved comfort, health, and reduced particulate emissions [19,20]. Despite these benefits, adoption of energy renovations remains slow, partly due to financial constraints [21], lack of technical skills [22], paucity of information on energy efficiency and renovating [23], and non-monetary costs such as time, effort and the hassle of having a construction team working in their home [24]. Uptake is also uneven across socioeconomic groups, with wealthier households more likely to adopt green technologies [25].

Existing research indicates that heating-system preferences in detached houses are shaped by a complex set of factors — including occupant characteristics, housing attributes, geographic location, economic considerations, and perceptions of system performance [see e.g. 26,27]. Far less is known about why households switch heating systems, although there is evidence that the probability of replacing systems increases with the lifetime of the equipment [28] and when occupancy changes [29], suggesting that timing constraints can limit who is in a position to take advantage of available incentives. The literature also shows that other renovation actions such as installing solar panels as well as the characteristics of both the old and the newly adopted heating system make a difference for replacement decisions [e.g., 30,31]. Perceptions of comfort, convenience, and overall functionality also interact with financial considerations, reinforcing patterns of technological lock-in that make some households more inclined to pursue renovations than others [32].

A well-designed subsidy programme could, in theory, address both the overall slow adoption of green technologies and the uneven distribution of their benefits. Yet existing evidence from green technology investment subsidies suggests that they finance measures that would have taken place even without the subsidy, indicating that efficiency gains are often modest [see e.g. 33,34]. In this context, the redistributive function of subsidies becomes particularly important: if subsidies do not significantly increase energy-saving investments, their main

value may lie in addressing the uneven financial burdens of renovations across households.

In practice, however, this redistributive potential is often unrealised. Studies consistently show that subsidy uptake is heavily skewed toward wealthier households. Allcott et al. [11] show that tax credits, utility-led efficiency programmes, and subsidised hybrid vehicles are primarily utilised by affluent "environmental homeowners", and Egner et al. [12] report that Norwegian retrofit subsidies largely accrue to high-income households. Comparable patterns are observed in Estonia [13,35], Australia [14], and Czechia [15], where uptake is lower in regions with fewer economic resources.

A key potential explanation for these disparities is self-selection: in the absence of targeting, households decide for themselves whether to participate in subsidy programmes, and this choice can be strongly influenced by factors such as income, social capital, and knowledge about energy efficiency. Hartman [36] was among the first to highlight that this selection would upwardly bias estimates of energy conservation programme effectiveness, as participants were more likely to conserve energy even without the programme. Self-selection has since been examined in the context of residential energy efficiency programme effectiveness [see e.g. 37]. The distributional implications of self-selection have received less systematic attention, even though the same mechanisms that drive self-selection – income, information, and the capacity to navigate the application process – vary across socioeconomic groups, suggesting that voluntary uptake may systematically favour better-off households. Accessing subsidies often requires the ability to plan renovations, identify available programmes, and select suitable contractors [38] — challenges that could be greater for less well-off groups. At the same time, information often does not reach all socioeconomic groups evenly, leaving many of the most vulnerable individuals least aware of support schemes [39]. As a result, subsidies may disproportionately benefit households that are already better off, potentially exacerbating inequality.

Awareness and understanding of a subsidy may, in turn, depend on the form the subsidy takes. Exploiting within-state variation in the form and generosity of incentives for hybrid vehicle adoption in the U.S., Gallagher and Muehlegger [40] find that conditional on incentive value, sales tax waivers generate a ten-fold greater impact on hybrid vehicle sales than income tax credits. They attribute this to poor consumer awareness and understanding of the credit rather than consumers discounting its value due to the lag between purchase and receipt of the credit. The authors conclude that the form of an incentive can matter as much as its generosity.

3. Institutional background

Practically all Finnish residential dwellings are equipped with heating systems, with hybrid combinations of multiple sources being widespread [41]. Wood, electricity, and ambient energy from heat pumps are the most common heating sources in Finnish detached and semi-detached houses (from now on referred to as detached houses) [42]. Oil heating, while now the fifth most common source overall, remains particularly prevalent in detached houses built between the late 1950s and 1970s [42].[2] Finland has a target of phasing out oil heating in all buildings by 2030 [43]. Thus, even though oil heating has decreased significantly, policy measures have been introduced to finalise the phase-out of oil heating.

To implement the policy target of phasing out oil heating by 2030, Finland introduced three subsidy programmes between 2020 and 2022 targeted at replacing oil heating with cleaner alternatives. The subsidies were available to residential property owners, and most applicants were detached house owners, as oil heating is most common in these buildings. Importantly, the three programmes were mutually exclusive, so households could apply to multiple subsidies but could receive only one. The subsidies, like most energy renovation subsidies, were aimed at homeowners rather than renters because homeowners are the ones making decisions regarding home improvements. For all three subsidies, the application could be completed online or mailed in paper form. Detailed information on each subsidy programme is provided in Table A.4.

The first subsidy — *cost-share subsidy* — was introduced in 2020, and it covered a share of the total renovation costs. The application required renovation plans, cost estimates and energy efficiency calculations to demonstrate that the planned renovation would significantly improve the building's energy performance class. The applicants had to hire a professional to perform the energy calculations, which created an additional cost of 300–400€ [44]. The subsidy calculation formula for the cost-share programme covered 100% of oil heating replacement costs but applied only to half of the measure's total costs (this could have been expressed simply as covering 50% of the total costs). The overall subsidy had a maximum threshold (4000€ per apartment for standard renovations, 6000€ for nearly zero emissions (NZEB) renovations). The cost-share subsidy required pre-approval before the renovation could begin, and the subsidy was paid after the renovation was completed upon submission of invoices and other supporting documents. The cost-share subsidy also covered other energy efficiency measures, such as solar panel installations, insulation and water-saving measures. However, we restrict our analysis to applicants who used the subsidy for an oil-heating retrofit.

The second subsidy — *lump-sum subsidy* — was introduced later in 2020. This was a subsidy of 4000€ when transitioning from oil heating to geothermal heating, an air-to-water heat pump or district heating, or 2500€ when transitioning to any other clean energy source.[3] The application form only required a description of the planned heating system replacement and a cost estimate. Pre-approval was recommended for the lump-sum subsidy before oil boiler replacement could begin, but it was not mandatory. Similar to the cost-share subsidy, the lump-sum subsidy was also paid after the renovation was completed.

The third subsidy — *tax credit* — became available in 2022, and it was an increased household tax credit for replacing oil heating with a clean alternative. The household tax credit is an existing instrument that is available for other household expenses such as cleaning, child care, or renovation measures. The tax credit only applies to labour costs and reduces the amount of income tax owed for the year. The increased tax credit for oil-heating retrofits covered 60% of labour costs from hiring a firm and 30% from hiring an individual contractor. However, the subsidy calculation could easily become more complicated if the applicant was also using tax credits for other household expenses or had insufficient taxable income to deduct the full tax credit. Both spouses in a household could receive the tax deduction for the same renovation, meaning that couples[4] were eligible for double the value compared to households with one taxpayer. To qualify, applicants needed to submit invoices detailing the labour costs for the replacement. The maximum sum of the tax credit subsidy was 3500€ for one person or 7000€ for couples. The application process for the tax credits is similar to any other tax deduction. Unlike the other two subsidies, information on the tax credit is provided in multiple languages and the application can be filled in using English.

[2] Contrary to many European countries, gas heating is very uncommon in Finland.

[3] In 2022, the subsidy also became available for switching away from gas heating.

[4] In tax law, a couple is defined as a married couple or a cohabiting couple that has previously been married or has a child together.

Table 1
Subsidy applicant numbers by year.

	2020	2021	2022	Total
Cost-share subsidy	634	220	193	1047
Lump-sum subsidy	8665	9681	8239	26,585
Tax credit	–	–	7734	7734

4. Data and methods

4.1. Data

4.1.1. Sample

We leverage nationwide[5] administrative data that allow us to examine the socioeconomic distribution of the subsidies at the household level. We have household-level data comprising the full Finnish population on: (1) subsidy applicants, (2) remaining oil heaters, (3) owners of detached houses with any heating system, and (4) earlier renovators. Group (1) are households, who applied for a subsidy for oil-heating phaseout between 2020–2022. Group (2) are owners of detached houses with oil heating who did not apply for any of the subsidies by the end of our sample period and still had oil heating listed as their main heating system in 2022 in the building stock registry. Group (3) is the entire population of detached house owners in Finland. Group (4) are a sample of detached house owners who switched away from oil heating during 2015–2019, before the subsidies became available.

Table 1 shows the number of applicants to each of the three programmes we have data for by year. These counts include households multiple times if they applied to more than one programme. We focus on the period 2020–22 because this was the original eligibility period for the cost-share and lump-sum subsidies before they were extended, and it is the period for which we have data on all three subsidies.

4.1.2. Data sources

Subsidy records. Data on all applicants during 2020–2022, including their renovation measures and costs, are obtained from the authority responsible for each programme (see Appendix A.1).

Finnish Building Stock Registry. The national building stock registry (RHR) contains information on the existing building stock, including construction year and main heating system. Remaining oil heaters (Group 2) are identified as detached-house owners whose building had oil heating recorded as its main system in the registry in 2022.[6] Earlier renovators (Group 4) are identified by observing buildings where the main heating system changed from oil to district heating, electric heating, or geothermal heating between 2015 and 2019, and then matching each building to its owner in the year the switch occurred.

Statistics Finland administrative data. Statistics Finland FOLK Basic Data and FOLK Household-dwelling unit data contain annual individual- and household-level information, respectively, for the full Finnish population. These data include household composition, the age of the household's reference person,[7] the highest education level and native language among household members, residence location type, home tenure type and disposable income. Common identifiers allow us to link individuals in groups 1–4 to the FOLK datasets. Statistics Finland's INFRA Spatial data module links all individuals in Finland to their residence building, enabling us to merge building characteristics from the RHR.

Three additional Statistics Finland modules are used to determine household tax status for the tax credit calculation. Taxable income is obtained from the FOLK Income module, which contains annual tax information for all individuals. Marital status is obtained from the FOLK_ASLII module on cohabitating couples, and whether the couple has children is obtained from the FOLK child-parent module. These are combined to calculate the counterfactual tax credit each household would have received in the absence of the oil-heating replacement deduction.

Energy Performance Certificate Registry. As a robustness check, the subsidy applicants are also compared with oil heaters in the Energy Performance Certificate Registry, which contains all buildings in Finland with energy performance certificates granted between 2015–2022.

4.1.3. Variable construction

Subsidy application. While households could apply for multiple subsidies, they could be granted only one. When we show results for the full group of subsidy applicants, we keep one observation per household. When comparing applicants to the three subsidies, we exclude households that applied to more than one programme.

Subsidy amount. In the lump-sum and cost-share subsidy data, the reported subsidy pertains to the whole renovation project; in the tax credit data, it pertains to the household's total annual tax credit. Because we are interested specifically in the subsidy granted for oil-heating replacement, we compute this component separately based on reported costs. For the lump-sum subsidy, the two quantities are equal by design. For the cost-share subsidy, the oil-heating component equals 50% of the oil-heating replacement costs, up to the maximum of 4000/6000€ per renovation. For the tax credit, we first calculate the oil-heating replacement deduction from the reported labour costs. The subsidy is then the difference between the tax credit actually paid and the counterfactual credit the household would have received without the oil-heating replacement deduction. The counterfactual tax credit is calculated using each household's taxable income from Statistics Finland's Income module, together with couples' tax status based on marital status and dependent children.

Disposable income. Household disposable income is equivalised by dividing household income by the number of equivalised adults, using the OECD equivalence scale. Income deciles are available as a ready-made variable in the administrative data; each household's decile is based on the distribution of equivalised disposable income in the full Finnish population in each year.

Education. Education level is categorised as: (1) upper secondary or lower (ISCED levels 0–3); (2) post-secondary non-tertiary (ISCED levels 4–5); and (3) higher education (ISCED levels 6–8).

Language. Finland has two national languages, Finnish and Swedish. We classify each household's language based on the most common registered native language among household members, distinguishing Finnish-speaking, Swedish-speaking, and foreign-language-speaking households, where the last group comprises households whose predominant native language is neither Finnish nor Swedish.

Residence area type. We classify residence areas using the Finnish urban–rural classification, which assigns each nationwide 250-by-250 metre grid cell to one of seven classes. We aggregate these into three categories: urban (inner and outer urban areas), peri-urban (peri-urban areas, local centres in rural areas, and rural areas close to urban areas), and rural (rural heartland areas and sparsely populated rural areas).

Timing. For applicant households, all background characteristics are measured in the year in which they apply for the subsidy. For earlier renovators, characteristics are taken from the year (between 2015–2019) in which the building stock registry records their heating system switching away from oil. For remaining oil heaters and for detached-house owners, characteristics are measured in 2019–2022, depending on the analysis, with the exact year indicated in the figure or table notes. All monetary values are expressed in 2022 values. Summary statistics for variables are presented in Section 5.1.

[5] We exclude the autonomous region of Åland, where the lump-sum and cost-share subsidies were not available.

[6] The building stock registry is updated based on construction permits/announcements, meaning that some heating system updates are not captured (see Appendix A.4 for more information). As a robustness check, the subsidy applicants are also compared with oil heaters in the Energy Performance Certificate Registry in Appendix A.4.

[7] A household's reference person is the household member with the highest disposable income.

4.2. Methods

We study which households apply for heating-system retrofit subsidies and, conditional on applying, which subsidy instrument they choose. We estimate two discrete choice models: a probit for the participation decision and a multinomial probit for the instrument choice. Discrete choice models have been widely used to study residential heating technology choice [see e.g. 27,30,45], and are well suited to our setting as they recover how household characteristics such as income, education, and housing characteristics shift the probability of each outcome and allow us to identify which household types self-select into subsidy programmes.

The models in both stages are grounded in the random utility framework of McFadden [46]. Household i derives utility from alternative j consisting of an observable, deterministic component, $V_{ij} = \mathbf{x}_i'\boldsymbol{\beta}_j$ and a random, unobservable component ε_{ij}:

$$U_{ij} = V_{ij} + \varepsilon_{ij}$$

The household chooses j if $U_{ij} > U_{im}$ for all $m \neq j$. Letting $y_{ij} = 1$ if household i chooses alternative j and 0 otherwise, the choice probability is:

$$\begin{aligned} E(y_{ij}) = \Pr(y_{ij}=1) &= \Pr\left(U_{ij} > U_{im} \ \ \forall m \neq j\right) \\ &= \Pr\left(V_{ij} + \varepsilon_{ij} > V_{im} + \varepsilon_{im} \ \ \forall m \neq j\right) \\ &= \Pr\left(\varepsilon_{im} - \varepsilon_{ij} < V_{ij} - V_{im} \ \ \forall m \neq j\right) \\ &= \Pr\left(\varepsilon_{im} - \varepsilon_{ij} < \mathbf{x}_i'(\boldsymbol{\beta}_j - \boldsymbol{\beta}_m) \ \ \forall m \neq j\right) \end{aligned}$$

Assuming the errors are normally distributed yields the stage-specific models below.

The two stages are estimated separately. Because the cost-share subsidy became available in early 2020, the lump-sum subsidy later that year, and the tax credit only in 2022, the first stage is estimated over the full 2020–2022 panel while the second is restricted to 2022, the first year all three instruments were available simultaneously. For the second stage, we adopt a multinomial model, as our interest lies in how household characteristics shape selection across programmes rather than in how attributes of the alternatives drive choice. We estimate a multinomial probit with independent errors, as a richer error covariance structure is not identified in our setting. As a robustness check, we re-estimate the model as pairwise binary probits and obtain consistent results (see Appendix A.3).

Stage 1: participation decision. The first stage examines whether household i applies for any subsidy in a given year t.[8] With two alternatives, $j \in \{0,1\}$, y_{ij} simplifies to the binary indicator $y_i = 1$ if household i applies and 0 otherwise. We estimate a probit model[9]:

$$E(y_i) = \Pr(y_i = 1) = \Phi\left(\mathbf{x}_i'\boldsymbol{\beta}\right) \tag{1}$$

The vector $\mathbf{x}_i$ contains annually measured household characteristics — equivalised disposable income, highest educational attainment, registered native language, years lived in the building, number of children, the reference person's age and sole-adult status, and the building's residential area type and age — and year indicators, which absorb aggregate differences across years so that applicants and non-applicants are compared within each year. Households remain in the sample until the year of their application, after which they exit; standard errors are clustered at the household level to account for repeated observations of the same household.

Stage 2: instrument choice. The second stage examines which subsidy instrument household i chooses, conditional on having applied in 2022. Here $j \in \{1,2,3\}$ denotes the cost-share subsidy, lump-sum subsidy and tax credit respectively, so that $y_{ij} = 1$ if household i chooses instrument j and 0 otherwise. We estimate a multinomial probit model:

$$\Pr(y_{ij}=1) = \Pr\left(\varepsilon_{im} - \varepsilon_{ij} < \mathbf{x}_i'(\boldsymbol{\beta}_j - \boldsymbol{\beta}_m) \ \ \forall m \neq j\right) \tag{2}$$

where $\mathbf{x}_i$ is the vector of household characteristics described above (with area dummies collapsed to a single urban indicator, and renovation costs added), and $\boldsymbol{\beta}_j$ is an alternative-specific coefficient vector for $j \in \{1,2,3\}$. The model is identified by normalising the coefficients of one alternative to zero, with the error variance fixed by the assumption of independent standard normal errors.

Reporting results. The probability of applying for any subsidy (Stage 1) and the probabilities of selecting each of the three subsidy instruments (Stage 2) have very different baseline levels, making absolute changes in probability difficult to compare. We therefore report all results as average semi-elasticities: average changes in the log predicted probability of the outcome for a one-unit increase in a continuous covariate, or when switching from 0 to 1 for a binary covariate.[10] In the first stage, this is the percentage change in the probability of applying for a subsidy; in the second stage, it is the percentage change in the probability of selecting each subsidy instrument.

5. Results

5.1. Baseline summary statistics and benefit distribution

We begin by examining the baseline situation before the subsidy programmes and analysing whether oil-heating households — the group targeted by the subsidies — differ from other owners of detached homes. To contextualise these differences and to explore observable characteristics linked to earlier renovation behaviour, we also introduce a third group: earlier renovators, meaning households that had already replaced their oil heaters prior to the subsidy programme. Table 2 presents summary statistics for detached-house owners, all oil-heating households, and earlier renovators in 2019.

Compared with other owners of detached houses, oil-heating households tend to be more disadvantaged: they are typically older, live in older buildings, and have lower equivalised disposable incomes. They are also less likely to have completed higher education. In contrast, earlier renovators appear significantly better off. They tend to be younger, more highly educated, and more likely to live in newer buildings in urban areas than the remaining oil-heating households.

The language composition of the sample is similar across groups: Finnish-speaking households account for 88%–91%, Swedish-speaking households for 7%–9%, and foreign-language households for 3%–4% of each group. However, these shares differ from the population at large. In 2019, Swedish-speakers accounted for approximately 5% of the population but are somewhat overrepresented among detached-house owners, reflecting the group's concentration in coastal and rural municipalities, where detached housing predominates, and its comparatively high homeownership rate. Foreign-language speakers, in turn, accounted for approximately 8% of the population but are strongly underrepresented, reflecting the group's lower homeownership rate.

Since the heating-system retrofit subsidies for oil heating were available to owners of oil-heated houses without income-based eligibility criteria, their distributional incidence depends on who takes them up. Fig. 1 compares the distribution of subsidy euros with that of potential recipients – owners of detached houses with oil heating – across income quintiles. Quintiles are defined over the full Finnish household population within each year.

[8] To simplify notation, year subscript t is dropped.

[9] In the probit model $\boldsymbol{\beta} \equiv \boldsymbol{\beta}_1 - \boldsymbol{\beta}_0$, since only the difference in utility between applying and not applying is identified, and the variance of the error difference $\varepsilon_{i0} - \varepsilon_{i1}$ is normalised to one.

[10] For continuous covariates $\partial \ln \Pr(y_{ij} = 1 \mid \mathbf{x}_i)/\partial x_{ik}$; for binary covariates, the discrete analogue $\ln \Pr(y_{ij} = 1 \mid x_{ik} = 1) - \ln \Pr(y_{ij} = 1 \mid x_{ik} = 0)$.

Table 2
Baseline summary statistics.
Notes: (1) The homeowners group consists of all owners of detached houses in 2019. (2) The oil heaters group consists of remaining oil heaters in 2019. (3) The earlier renovators group consists of households that switched away from oil heating during 2015–2019. "Diff." columns report pairwise differences in means. Standard deviations in parentheses.

	(1) Homeowners	(2) Oil heaters	(3) Earlier renovators	(4) Diff. (1)–(2)	(5) Diff. (3)–(2)
Reference person age	51.66	58.77	52.73	−7.11***	−6.04***
	(14.77)	(16.41)	(16.45)	(0.04)	(0.19)
Finnish-speaking[a]	0.91	0.88	0.89	0.02***	0.01
	(0.29)	(0.32)	(0.31)	(0.00)	(0.00)
Swedish-speaking[a]	0.07	0.09	0.07	−0.02***	−0.02***
	(0.25)	(0.29)	(0.26)	(0.00)	(0.00)
Foreign-language-speaking[a]	0.03	0.03	0.04	0.00	0.01***
	(0.16)	(0.16)	(0.19)	(0.00)	(0.00)
Building age	40.40	52.29	51.49	−11.89***	−0.80***
	(27.21)	(19.11)	(19.75)	(0.05)	(0.23)
Urban residence area[a]	0.38	0.39	0.52	−0.00	0.13***
	(0.49)	(0.49)	(0.50)	(0.00)	(0.01)
Higher education[a]	0.46	0.33	0.47	0.13***	0.14***
	(0.50)	(0.47)	(0.50)	(0.00)	(0.01)
Disposable income	38 462.71	36 568.94	42 021.97	1893.77***	5453.04***
	(51 700.19)	(46 557.77)	(90 232.04)	(107.30)	(1026.77)
1st income quintile[a]	0.11	0.14	0.06	−0.03***	−0.08***
	(0.31)	(0.35)	(0.24)	(0.00)	(0.00)
Observations	2,482,504	207,701	11,875	2,690,205	219,576

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
[a] Shares of all households in groups.

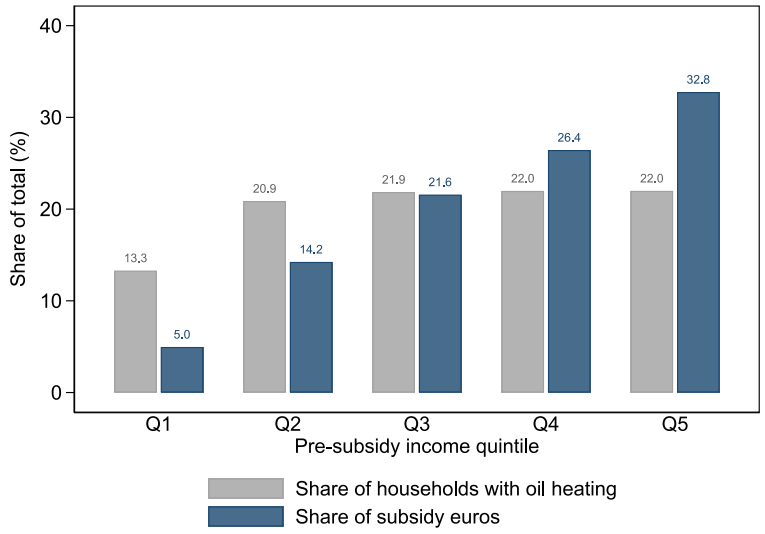


Fig. 1. Distribution of oil heaters and heating-system retrofit subsidies among owners of detached houses pooled over 2020–2022.
Notes: Grey bars show the share of owners of detached houses with oil heating falling in each income quintile; navy bars show the corresponding distribution of total subsidy euros.

The grey bars show how owners of detached houses with oil heating are distributed across the quintiles. Oil-heating homeowners are underrepresented in the lowest quintile (13.3%) and spread roughly evenly (21%–22%) across the four higher quintiles – a pattern consistent with rising homeownership and declining oil-heating prevalence roughly offsetting each other as income increases. The navy bars show how the subsidy euros paid to this group in 2020–2022 are distributed across the same quintiles. If subsidy euros had been allocated in proportion to the potential recipients, the navy bar would match the grey bar in every quintile.

The two distributions instead diverge markedly. The lowest quintile received only 5% of subsidy euros — well under half its proportional share. The share of subsidy euros then rises with income, nearly reaching the share of potential recipients in the third quintile and clearly surpassing it in the two highest quintiles. The subsidy thus disproportionately accrued to higher-income households within its target group. To better understand this distributional outcome, we next examine what types of households apply for the subsidies.

5.2. Who applies for a subsidy?

Fig. 2 shows the income distribution for the subsidy applicants and comparison groups. We see that the income distribution for remaining oil heaters is flatter than that for average detached house owners, consistent with our earlier observation that oil heaters are on average less

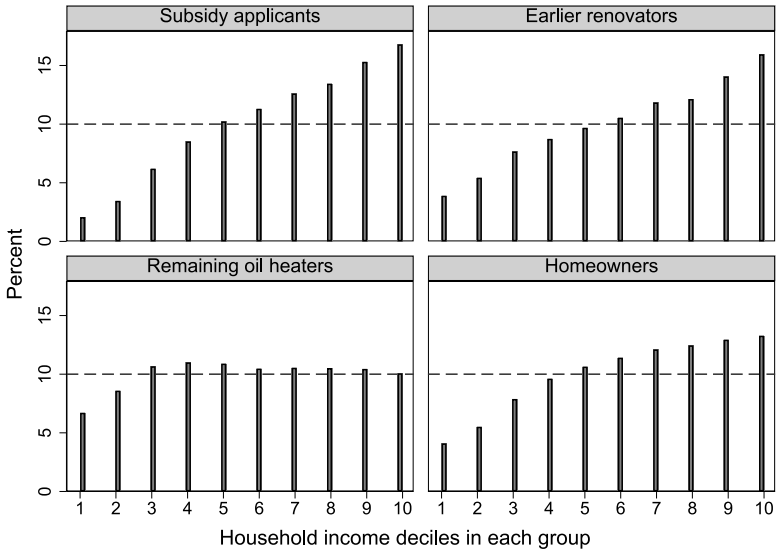


Fig. 2. Distribution of households across income deciles by group
Notes: The bars show the share of households in each income decile within four groups: subsidy applicants (households that applied for heating-system retrofit subsidies for oil heating during 2020–2022), earlier renovators (households that switched away from oil heating during 2015–2019), remaining oil heaters (owners of detached houses who still used oil heating as their main system in 2022 and had not applied for any subsidy), and homeowners (all owners of detached houses in 2022).

well-off than other homeowners. By comparison, the distribution for subsidy applicants is considerably more skewed to the right, indicating that subsidy applicants disproportionately come from the top of the income distribution. The distribution for subsidy applicants appears to be equally – if not more – regressive as that of earlier renovators, suggesting that the subsidies have not been effective at broadening the pool of households replacing oil heating.

Fig. 3 shows selected estimates from a probit regression comparing subsidy applicants with remaining oil heaters. The full results are provided in Table A.5. The results represent a percentage change in the predicted probability of applying for a subsidy associated with a one-unit change in continuous variables or a change from 0 to 1 for categorical variables. We see that subsidy applicants tend to have higher equivalised disposable income than the remaining oil heaters. This relationship between income and the probability of applying appears to be relatively linear, as demonstrated in specification (5) of Table A.5 showing the estimates for each income quintile. In addition to having higher income, subsidy applicants tend to be younger, more urban and more highly educated. They are more likely to live in newer houses, in which they have been living in for a shorter period of time, and are less likely to be renting out their house. They are also less likely to be a single-person/parent household or to be Swedish- or foreign-language-speaking households. Table A.5 in the Appendix provides the full table of results. The results from the robustness test in Appendix A.4 are very consistent with our main results.

5.3. Choice of subsidy

While the subsidies overall tend to have been very unequally distributed, we see differences across the three programmes. Table 3 provides summary statistics and Fig. 4 income distributions for applicants of the three subsidies. We see that the average equivalised disposable income among applicants is lower for the lump-sum subsidy than for the other two subsidies, and its income distribution is flatter, indicating that this subsidy has attracted more low- and middle-income applicants.

Fig. 5 provides selected estimates from the multinomial probit regression comparing applicants of the three subsidies in 2022. Full results are provided in Table A.6. The results represent a percentage change associated with a one-unit change in the continuous variables or a change from 0 to 1 for the categorical variables. The results confirm the observation made above that average equivalised disposable income among applicants is lower for the lump-sum subsidy than for the other two subsidies. Specification (5) in Table A.6 shows the estimates for each income quintile and demonstrates that the gap between the probability of applying for the lump-sum subsidy as opposed to the tax credit widens linearly with each income quintile.

We see that, on average, the applicants to the tax credit tend to be older and have higher income. This subsidy was also more likely to attract foreign-language-speaking households and less likely to attract single-person/parent households. The cost-share subsidy had the fewest applicants overall, so most of the estimates are not statistically significant and should therefore be interpreted with caution. However, the applicants in this subsidy were clearly the most urban, lived in the newest buildings and were more likely to be Swedish-speaking. Applicants in the cost-share and tax credit subsidies tended to be the most highly educated. The lump-sum subsidy, meanwhile, averaged younger, less urban applicants with lower equivalised disposable incomes. Lump-sum subsidy applicants also tended to have lower renovation costs than applicants in the other subsidies.

5.4. Potential explanations

In general, we observe that applicants of heating-system retrofit subsidies tend, on average, to be significantly better off than remaining oil heaters. A potential explanation for this is the high upfront investment required for the renovation. As shown in Table 3, the renovation costs for replacing oil heating are in the realm of 15 000–20 000€ . Although the subsidy covered a significant share of the renovation costs, averaging approximately 30%, the applicant still had to bear the majority of the costs. These remaining renovation costs made up a significant

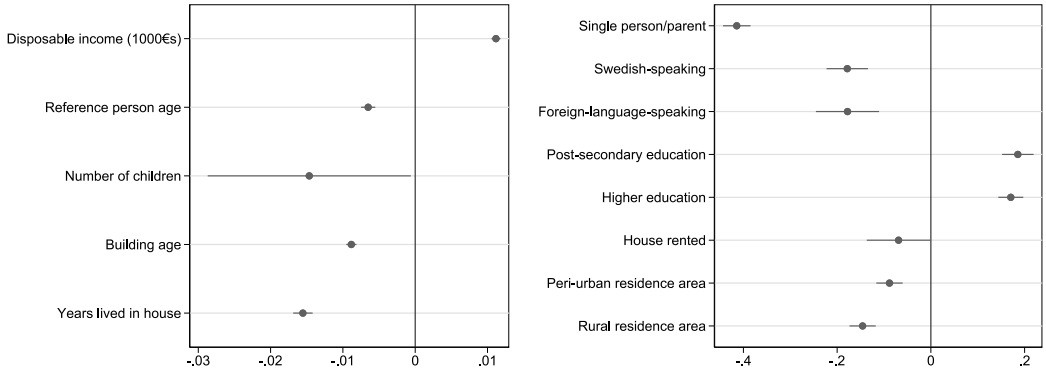


Fig. 3. Probit estimates of the association between household characteristics and the probability of applying for a subsidy.
Notes: Selected point estimates and 95% confidence intervals from probit specification (4) in Table A.5. Estimates are average semi-elasticities (percentage changes in the predicted probability of applying). 'Finnish-speaking' is the omitted language category. 'Upper secondary or lower education' is the omitted education level category. 'Urban residence area' is the omitted residence location category.

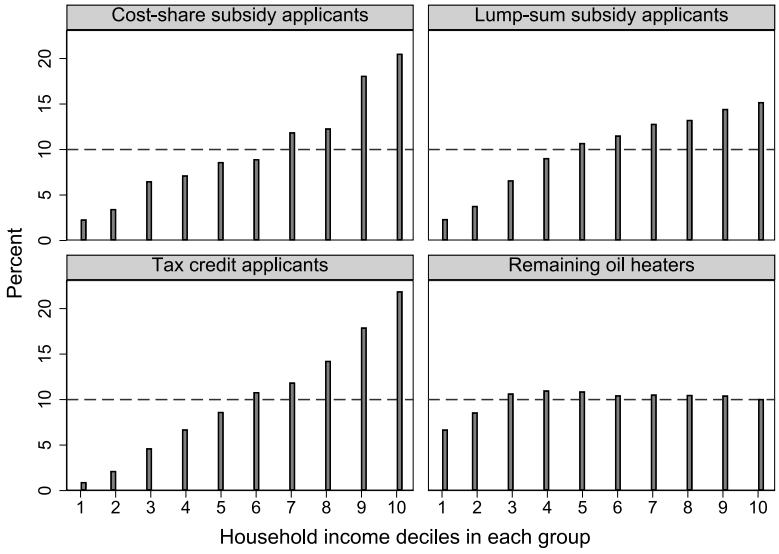


Fig. 4. Distribution of households across income deciles by subsidy programme.
Notes: The bars show the share of households in each income decile for applicants to three subsidy programmes, and for remaining oil heaters (owners of detached houses who still used oil heating as their main system in 2022 and had not applied for any subsidy).

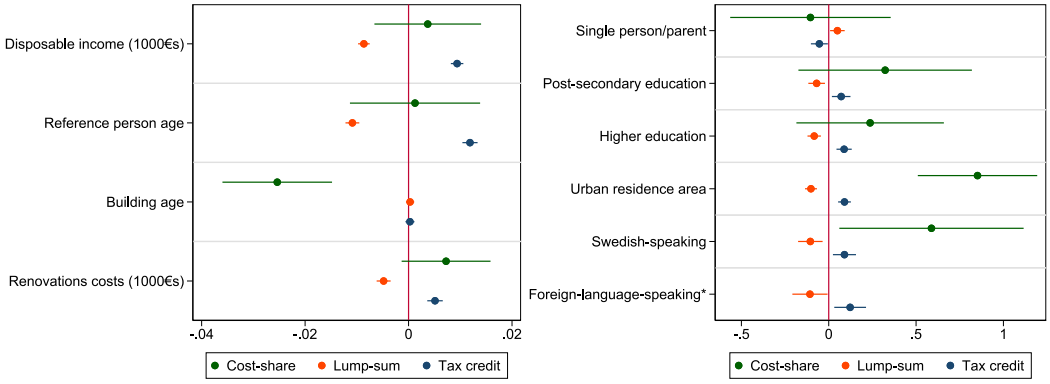


Fig. 5. Multinomial probit estimates of the association between household characteristics and the probability of selecting each subsidy instrument.
Notes: Selected point estimates and 95% confidence intervals from multinomial probit specification (4) in Table A.6. Estimates are average semi-elasticities (percentage changes in the predicted probability of selecting each instrument). *The coefficient on the cost-share instrument for foreign-language-speaking households is large but imprecisely estimated and is omitted here for readability; see Table A.6. 'Finnish-speaking' is the omitted language category. 'Upper secondary or lower education' is the omitted education level category. The omitted residence area category is 'non-urban residence areas'.

Table 3
Summary statistics for applicants to the oil phase-out subsidies during 2020–2022.
Notes: "Diff." columns report pairwise differences in means. Standard deviations in parentheses.

	(1) Cost-share	(2) Lump-sum	(3) Tax credit	(4) Diff. (2)–(1)	(5) Diff. (3)–(1)
Reference person age	52.77	54.72	58.72	1.95***	5.96***
	(15.44)	(15.41)	(14.84)	(0.54)	(0.56)
Building age	48.56	50.95	51.13	2.39**	2.57**
	(22.10)	(20.08)	(22.28)	(0.77)	(0.81)
Urban residence area[a]	0.50	0.42	0.47	−0.08***	−0.03
	(0.50)	(0.49)	(0.50)	(0.02)	(0.02)
Higher education[a]	0.52	0.42	0.48	−0.10***	−0.04
	(0.50)	(0.49)	(0.50)	(0.02)	(0.02)
Disposable income	43 883.35	40 209.67	45 973.86	−3673.68***	2090.51
	(27 054.68)	(29 088.60)	(107 361.78)	(944.14)	(1652.23)
Renovation costs	18 053.29	15 731.69	17 531.55	−2321.60***	−521.74
	(8290.02)	(21 747.92)	(16 902.15)	(317.87)	(355.92)
Subsidy level	5153.26	4008.73	3183.80	−1144.53***	−1969.46***
	(2947.45)	(1345.49)	(2440.75)	(101.08)	(105.39)
Observations	857	22,682	6145	23,539	7002

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
[a] Household shares per group.

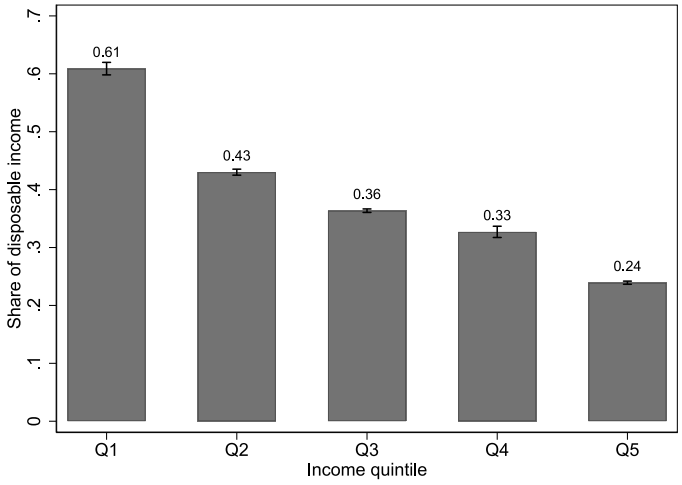


Fig. 6. Renovation costs as a share of income after subsidy.
Notes: Average post-subsidy renovation costs as a share of households' equivalised annual disposable income by income quintile.

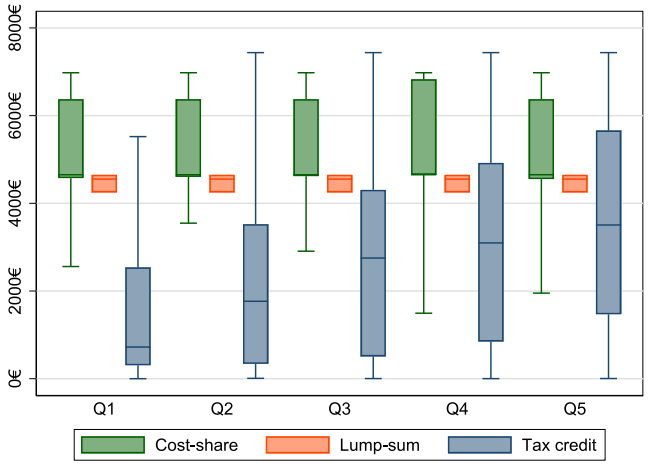


Fig. 7. Average subsidy level in euros by income quintile.
Notes: Subsidy levels in euros, adjusted for inflation.

share of a household's annual equivalised disposable income, especially for lower-income households. As shown in Fig. 6, the share increases significantly from an average of 24% in the highest-income quintile to 61% in the lowest-income quintile.[11]

However, the distributions of the three subsidies differ. One of the key reasons for the attractiveness of the increased tax credit, especially for wealthy households and households with more than one taxpayer, is that it can potentially offer the largest subsidy levels: 7000€ per household. As the tax credits are tax deductions, there are more households in lower-income groups with insufficient taxable income to claim the maximum deduction. Furthermore, both spouses in a household can apply for the tax credit, meaning that couples could receive double the amount compared with single-taxpayer households.[12] Fig. 7 shows the average granted subsidies by income quintile. We see that the distribution of subsidy levels for the tax credit is very regressive, increasing substantially with income. By comparison, the cost-share and lump-sum subsidies did not depend on income, resulting in relatively flat average subsidy levels across the income distribution. Although the tax credit could potentially provide the highest level of support, its average subsidy levels remained below those of the other two instruments: the credit only applied to labour costs, and claiming the maximum amount required a sufficiently high taxable income.

6. Discussion

Our results indicate that the heating-system retrofit subsidies for replacing oil heating have predominantly benefited higher-income households, who also tend to be younger, more urban and more highly educated than other oil heaters. They are more likely to live in newer houses, in which they have been living for a shorter period of time, and less likely to be renting out their house. While previous research has found energy renovation subsidies to primarily accrue to high-income households, our results are striking given that the subsidies

[11] The subsidy levels pertain specifically to the replacement of oil heating systems (see Section 4.1.3).
[12] Finnish tax law transfer rules also mean that it is enough if one of the adults has sufficient income to be able to reap the entire benefit.

were targeted specifically at households using oil heating, a group we find to be generally less well off than other owners of detached houses.

While high-income households tend to consume more energy [47, 48], which could be an argument in favour of incentivising their energy renovations first, there is reason to believe that many in this group would have undertaken the renovation even without a subsidy. Oil heating replacements in Finland can be cost-efficient, particularly when the existing system reaches the end of its life or when oil prices are high [49]. More well-off households have better means to make use of profitable investments even without the subsidy, limiting the cost-effectiveness of non-income-targeted instruments. Consistent with this, our results show that earlier renovators, who made the investments before the subsidies were available, were also predominantly high-income households.

For a subsidy to be just, it should neither add to existing vulnerabilities nor create new ones. Our results, however, show that the subsidies in 2020–2022 primarily attracted the most well-off households from the pool of oil heaters, leaving behind, on average, a more vulnerable group of remaining oil heaters. Overall, this group can be described as ageing, lower-income residents, who typically live in more rural areas, where housing values tend to be lower than in urban areas. These patterns raise concerns from both distributional and recognition justice perspectives and indicate that simply targeting a less-advantaged population through a specific renovation action is insufficient to address equity concerns when self-selection shapes who ultimately benefits.

Due to their steadily declining number in Finland, the remaining oil heaters are more vulnerable to shocks in heating oil prices and future supply availability, and they continue to suffer from higher variable costs associated with continued fossil fuel use. Following the revised Energy Performance of Buildings Directive (EPBD), ETS2, and the Finnish Government's target of phasing out oil heating by 2030, the remaining oil heaters will face increasing pressure to undertake energy renovations. Existing vulnerabilities may concurrently even be exacerbated for the last adopters – or those who decide not to adopt at all – as the cost of oil heating is expected to increase further following the new ETS2. As a more rural group, they are likely to simultaneously face a disproportionate burden from transport fuel price rises caused by ETS2, since they may be more dependent on cars due to longer travel distances and worse public transport services and may have fewer opportunities to buy an electric vehicle due to lower financial means.

Beyond income, several other factors also shape self-selection into the subsidy programmes, contributing to systematic differences between applicants and the remaining oil-heating households. For example, subsidy applicants are considerably less likely to be renting out their house, which could be indicative of the split incentive problem that has been identified in earlier research, where owners have less incentive to invest in energy efficiency because energy costs are paid by the tenant [50]. The applicants are also less likely to be Swedish- or foreign-language-speaking than Finnish-speaking. For the foreign-language-speaking group, a likely explanation is a language-barrier: only the tax credit provided information in multiple languages and could be applied for in English, and consistent with this, it was relatively more popular among foreign-language-speaking applicants. For the Swedish-speaking group, the application process itself is unlikely to have been an obstacle, as all three subsidies could be applied for in Swedish. Their lower application propensity may instead reflect less extensive outreach in Swedish-language media, or characteristics of the Swedish-speaking population that our controls do not fully capture.

It would be theoretically possible to design a subsidy in a manner that alleviates distributional concerns. The Social Climate Fund (SCF), which will use proceeds from ETS2, is one instrument that could be utilised for new policies, and the European Commission's guidance on Social Climate Plans [51] indicates that social impacts and challenges faced by vulnerable groups should be adequately addressed with targeted policies. Our research provides insights on several aspects regarding how countries can design fossil fuel phase-out policies in a more equitable manner.

Our results comparing three subsidy instruments complement Gallagher and Muehlegger [40], whose analysis of aggregate state-level hybrid sales suggests that instrument design and salience can matter as much as generosity. They find that a tax credit generates considerably smaller sales responses than an equally generous sales tax waiver — likely due to poor consumer awareness or understanding of the credits. By contrast, in our setting all three instruments are paid after the renovation, and the tax credit is a well-known existing instrument. Based on household-level data, we find that a tax credit was the most regressive instrument and least favourable to lower-income households. The regressivity is likely to stem from the requirement of sufficient taxable income to claim it. In addition to its built-in mechanism enabling larger gains for higher-income households, the Finnish increased tax credit also disadvantaged single-adult households, as couples with sufficiently high taxable income could jointly obtain a higher subsidy.

However, the tax credit was a relatively popular instrument among the three concurrently available subsidies. It attracted more older applicants and foreign-language-speaking applicants, potentially due to its familiarity as an existing instrument – available, for example, for elderly care services – and to the availability of information in various languages. These are important features of the tax credit that could be applied to other energy renovation subsidies.

The lump-sum subsidy was the most popular instrument overall, and the one favoured by households in the lower-income deciles. We believe this was due to its relatively straightforward application process and its easily comprehensible nature. It was also the most salient of the three instruments: a fixed, pre-announced amount that required no calculation of eligible costs or tax liability, leaving no uncertainty about the final payment at the time of the decision. This simplicity may explain its appeal among lower-income households, for whom uncertain or liability-dependent subsidies may carry greater risk.

Based on the small applicant numbers, the cost-share subsidy appears to have been the least preferred instrument among oil heaters.[13] It had the most complicated and expensive application process: it required energy-efficiency improvement calculations by a professional, which generated an additional cost. The subsidy also involved multiple rates for different measures and a priori uncertainty about whether the energy criteria would be met, making it less salient than the lump-sum subsidy, which yielded a comparable payment with greater certainty.

Based on the low subsidy take-up by lower-income households, more support for these groups may be needed in the future. One potential obstacle to address could be the difficulty in securing a renovation loan[14] if the applicant has low income or lives in an area

[13] We conducted a survey directed toward cost-share subsidy recipients in late 2021. It revealed concerns pertaining to the understanding of the subsidy rate and the difficult application process. These factors may have dissuaded potential applicants from the cost-share subsidy and made the lump-sum subsidy overall more appealing. Furthermore, survey respondents were critical of lengthy decision times and the subsequent uncertainty regarding their eligibility.

[14] As all three subsidies required the applicant to finance the entire renovation prior to getting the subsidised share back, we cannot assess, how the number or characteristics of applicants might have been if the full renovation costs – or even just the subsidy part – had been paid up-front. Based on our research, it is nevertheless likely that even a generous subsidy rate may not be sufficient to induce uptake in low-income groups if it is not coupled with a loan instrument.

with low property values. Furthermore, potential market-level impacts should be considered. Subsidies can increase the demand for energy renovations, which in turn can increase renovation prices, necessitating even larger upfront investments. Alleviating financial obstacles could require a higher subsidy rate for the lowest-income groups – potentially made possible by means-testing – and/or improved access to loans for energy renovations.

As the remaining oil heaters are an increasingly vulnerable group, future subsidies should be designed with this group in mind and accompanied by understandable information on the technology choices and their use, along with practical assistance for the applicants. It is important that the groups impacted by the policy have a voice in its design. Involvement and agency cannot only improve policy design but also increase policy acceptance [52].

Despite having access to exceptionally rich administrative data on Finnish households and buildings, our study has some limitations. Heating-system information in the building stock registry is updated mainly in connection with permit-requiring renovations. This means that some of the households classified as oil-heaters in the data may have already switched to another heating system. For the same reason, we do not observe oil-to-oil boiler replacements, nor details such as supplementary heating sources or the presence of solar panels. Moreover, the study period for which we have data covers only a few years of the subsidy programmes, so we cannot assess long-run outcomes, and because our second-stage model of instrument choice conditions on 2022 applicants, some caution is warranted in extending the estimates to the wider eligible population. Finally, our instrument choice model assumes independent errors across the alternatives, a restriction we cannot relax in our setting (see Appendix A.3).

As there may be some technology-specific features in the energy transition, future research should specifically examine the phase-out of gas heating, which is more common in other European countries, as well as the subsidy impacts from other renovation actions including building envelope measures and renewable generation investments.

7. Conclusions

Fossil fuel-based systems need to be urgently phased out in space heating and cooling. While this phase-out in the Finnish detached housing stock is well underway, there will be increasing pressure from the ETS2, on remaining oil heaters to transition away from oil. To avoid exacerbating existing inequalities, it is important to carefully design how the proceeds are distributed, for example, via the Social Climate Fund.

Subsidies can play a role in this transition, but different instruments can lead to various distributional outcomes. Our results show that, due to self-selection, a subsidy may be regressive even when it targets a specific technology commonly used by low- and middle-income households. This highlights that technology-based targeting does not guarantee an equitable outcome, and household-level targeting should be considered. When this is not possible, our results suggest that the least regressive distribution is likely to be achieved with a simple lump-sum instrument.

The remaining oil heating residential building stock in Finland is harder to abate. It appears that a combination of information and hands-on application support, government-backed loans and potentially a higher subsidy level – perhaps inversely proportional to the income level of the applicant – may be needed to incentivise these households to renovate. Naturally, our results are based on a comparison of subsidies that were available around the same time, meaning that the distribution could look different if only one subsidy were available. Further research is needed to explore the distributional effects under different subsidy instrument availabilities and features. However, this setting provided a fruitful opportunity to isolate the effects of a subsidy programme's policy design on its distributional outcomes, which will hopefully inform the planning of future energy renovation subsidies.

The phase-out of fossil fuel-based heating and cooling in the residential housing stock is subsidised to improve energy efficiency and reduce emissions. In Finland, as in many other countries without oil and natural gas reserves, the replacement of fossil fuel-based heating systems is also a question of national energy security, and this aspect has become a political priority especially since the beginning of the Russian invasion of Ukraine in 2022. While our study took place in Finland, the results can be applicable to other European countries' planning policies for speeding up the transition from gas heating to heat pumps. As a country that has made significant progress in phasing out fossil fuel-based heating, Finland offers valuable insights on the distributional effects of energy policies for other countries undergoing similar transitions. Incorporating these considerations at early stages of the transition is crucial to mitigate perceived injustices and reduce resistance, including populist opposition, to the energy transition.

CRediT authorship contribution statement

Selina Clarke: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Jouni Räihä:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Hanna-Liisa Kangas:** Writing – review & editing, Writing – original draft, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

A.1. Subsidy table

See Table A.4.

Table A.4
Information on subsidies.
Notes: *30% if individual contractor hired instead of firm. **The cost-share subsidy was also available for housing associations and ARA communities (non-profit entities within state-supported housing production), but these buildings less commonly have oil heating and are not analysed in this paper. *** Vacation homes are excluded from our analysis.

	Cost-share subsidy	Lump-sum subsidy	Tax credit
Time span	January 2020–2023	September 2020–2026	2022–2027
Eligible measures	21 energy efficiency measures (one of which is oil heating replacement) + planning costs	Oil or gas heating replacement	Oil heating replacement
Subsidy level	10%, 25% or 50% of the costs of each measure, but max. 4000€ (6000€) per apartment for a standard (NZEB) renovation.	4000€ when heating system replaced with geothermal, heat pump or district heating. 2500€ for other energy sources.	60%* of labour costs, but max. 3500€ per person (Note, this is the maximum for all tax credit deductions and the deductible is 100€ per person)
Who can apply	Owners of residential buildings, at least half of which are occupied year-round**	Owners of small residential buildings occupied year-round	Owners, or close relatives (spouse, children) of owner of permanent residence property or vacation home***
Eligibility requirements	Building must be lived-in throughout the year, a building's *E*-value must improve by 32%–44% or must fall below specific thresholds for NZEB renovations.	Building must be lived-in throughout the year. New heating system cannot use fossil fuels.	New heating system cannot use fossil fuels.
Application process	Apply before renovation. Must provide renovation plans, cost estimates and energy certificate/energy improvement calculations.	Must provide renovation plans, cost estimates	Apply after renovation. Report deduction in tax card/tax return.
Responsible authority	Housing Finance and Development Centre of Finland (ARA)	The Centre for Economic Development, Transport and Environment (ELY)	Tax Administration

A.2. Regression results

See Tables A.5 and A.6.

A.3. IIA

The Independence of Irrelevant Alternatives (IIA) assumption requires the relative odds of choosing between any two alternatives to be unaffected by the presence or attributes of other options. We test this assumption using the Hausman–McFadden test, comparing a multinomial logit tested on all three alternatives to one in which the tax credit alternative is excluded. The results, reported in Table A.7, reject IIA.

In principle, a multinomial probit with a freely estimated error covariance structure would relax this restriction. In our setting, however, the covariance parameters are not credibly identified: our model does not include alternative-specific covariates, and the cost-share alternative has a substantially smaller sample than the other two subsidies. We therefore estimate the multinomial probit with independent errors, which imposes a substitution structure similar to IIA.

To assess whether this restriction drives our results, we re-estimate the instrument choice as a set of pairwise binary probits (Table A.8). Each column compares two subsidy instruments on the subsample of 2022 applicants who chose one of the two, without imposing any structure on substitution with the third alternative. The results are consistent with the multinomial probit estimates, suggesting that our conclusions about which household characteristics shape instrument choice are not sensitive to the independence assumption.

A.4. Alternative data for oil heaters

The building stock registry is updated based on construction permits and voluntary construction announcements. This means that some heating system updates that do not require permits may not be updated and some of the households classified as oil heaters in the data may have already switched to another heating system. As a robustness check, the subsidy applicants are also compared with oil heating owners of detached houses in the Energy Performance Certificate (EPC) Registry during 2020–2022. The EPC Registry contains more up-to-date information on a building's heating system. However, an EPC is only required for existing buildings when the building is rented or sold, meaning that the data only contain a selected sample of oil heaters. This is demonstrated in Table A.9, which presents summary statistics for oil heaters and subsidy applicants with and without an EPC. We see that, on average, those with an EPC tend to be considerably younger, live in newer buildings, are more urban and more likely to have completed higher education than those without an EPC. In terms of income, subsidy applicants with an EPC tend to average a slightly lower income than those without an EPC. By contrast, oil heaters with an EPC tend to have slightly higher income than those without one. However, the difference is significant only at the 10% level. These results highlight the selectiveness of the samples in the EPC registry, and for this reason, we only compare EPC oil heaters with subsidy applicants with an EPC.

Column (2) in Table A.10 provides the results from the robustness test, where subsidy applicants with an EPC are compared with EPC oil heaters. For comparison, column (1) shows the main results from Table A.5 specification (4), where all subsidy applicants are compared with all oil heaters in the building stock registry. While the results differ slightly in terms of magnitude, as expected with a more selective sample, almost all signs are the same as in the main results, providing further support for our findings. The only exceptions are reference person's age and number of children: in one of the two estimations, the former is not statistically significant and the latter is significant only at the 10% level.

Data availability

Data used in this study originate from national authorities and, due to data protection rules, cannot be made publicly available. Interested researchers may request access from relevant authorities.

Table A.5
Full results from probit regression.
Notes: Estimates are average semi-elasticities (percentage changes in the predicted probability of applying). Omitted categories: 'Finnish-speaking', 'upper secondary or lower education', 'urban residence area' and 'income quintile 1'.

	(1)	(2)	(3)	(4)	(5)
Disposable income (1000€s)	0.013***	0.012***	0.012***	0.011***	
	(0.000)	(0.000)	(0.000)	(0.000)	
Reference person age	−0.012***	−0.013***	−0.006***	−0.007***	−0.003***
	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)
Single person/parent	−0.440***	−0.435***	−0.419***	−0.414***	−0.297***
	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)
Number of children	0.017*	0.006	−0.012	−0.015*	0.031***
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Swedish-speaking	−0.225***	−0.234***	−0.200***	−0.178***	−0.175***
	(0.022)	(0.022)	(0.022)	(0.023)	(0.023)
Foreign-language-speaking	−0.071*	−0.082*	−0.168***	−0.178***	−0.115***
	(0.033)	(0.033)	(0.034)	(0.034)	(0.034)
Post-secondary education		0.197***	0.193***	0.186***	0.134***
		(0.017)	(0.017)	(0.017)	(0.017)
Higher education		0.191***	0.184***	0.171***	0.150***
		(0.013)	(0.014)	(0.014)	(0.014)
Building age			−0.009***	−0.009***	−0.009***
			(0.000)	(0.000)	(0.000)
Years lived in house			−0.016***	−0.016***	−0.016***
			(0.001)	(0.001)	(0.001)
House rented			−0.069*	−0.069*	0.030
			(0.035)	(0.035)	(0.035)
Peri-urban residence area				−0.088***	−0.078***
				(0.014)	(0.014)
Rural residence area				−0.146***	−0.117***
				(0.014)	(0.014)
Income quintile 2					0.470***
					(0.029)
Income quintile 3					0.732***
					(0.028)
Income quintile 4					0.875***
					(0.029)
Income quintile 5					1.070***
					(0.029)
Observations	600,760	600,760	600,400	600,348	600,348
Pseudo-R2	.033	.034	.039	.04	.044
Log likelihood	−119,069	−118,948	−118,162	−118,098	−117,620
AIC	238,155	237,916	236,350	236,226	235,276
BIC	238,245	238,029	236,497	236,396	235,479
Prob > x2	<0.001	<0.001	<0.001	<0.001	<0.001
Year fixed-effects	Yes	Yes	Yes	Yes	Yes

Standard errors clustered at household level in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6
Full results from multinomial probit regression.
Notes: Estimates are average semi-elasticities (percentage changes in the predicted probability of selecting each instrument). Baseline: lump-sum subsidy applicants; omitted categories: 'Finnish-speaking', 'upper secondary or lower education', 'income quintile 1' and 'non-urban residence areas' (all other area types).

	(1)		(2)		(3)		(4)		(5)	
Disposable income (1000€s)										
Cost-share	0.011**	(0.004)	0.008	(0.005)	0.006	(0.005)	0.004	(0.005)		
Lump-sum	−0.010***	(0.001)	−0.009***	(0.001)	−0.009***	(0.001)	−0.009***	(0.001)		
Tax credit	0.011***	(0.001)	0.010***	(0.001)	0.010***	(0.001)	0.009***	(0.001)		
Reference person age										
Cost-share	0.003	(0.006)	0.000	(0.006)	0.000	(0.006)	0.001	(0.006)	0.000	(0.007)
Lump-sum	−0.011***	(0.001)	−0.011***	(0.001)	−0.011***	(0.001)	−0.011***	(0.001)	−0.013***	(0.001)
Tax credit	0.012***	(0.001)	0.012***	(0.001)	0.012***	(0.001)	0.012***	(0.001)	0.014***	(0.001)
Single person/parent										
Cost-share	−0.145	(0.217)	−0.153	(0.219)	−0.099	(0.229)	−0.105	(0.234)	−0.142	(0.245)
Lump-sum	0.069***	(0.021)	0.066**	(0.021)	0.050*	(0.021)	0.050*	(0.021)	0.008	(0.022)
Tax credit	−0.075**	(0.024)	−0.072**	(0.025)	−0.054*	(0.025)	−0.054*	(0.025)	−0.006	(0.025)
Number of children										
Cost-share	0.201*	(0.088)	0.175	(0.091)	0.156	(0.096)	0.149	(0.100)	0.137	(0.102)
Lump-sum	−0.081***	(0.011)	−0.074***	(0.011)	−0.069***	(0.011)	−0.067***	(0.011)	−0.093***	(0.011)
Tax credit	0.084***	(0.012)	0.077***	(0.012)	0.072***	(0.012)	0.070***	(0.012)	0.099***	(0.013)
Post-secondary education										
Cost-share			0.380	(0.243)	0.365	(0.249)	0.323	(0.253)	0.337	(0.258)
Lump-sum			−0.073**	(0.024)	−0.073**	(0.024)	−0.069**	(0.024)	−0.040	(0.024)
Tax credit			0.076**	(0.027)	0.076**	(0.027)	0.071**	(0.027)	0.039	(0.028)
Higher education										
Cost-share			0.396	(0.205)	0.349	(0.210)	0.237	(0.215)	0.247	(0.221)
Lump-sum			−0.100***	(0.019)	−0.095***	(0.019)	−0.083***	(0.020)	−0.073***	(0.020)
Tax credit			0.104***	(0.022)	0.099***	(0.022)	0.088***	(0.022)	0.076***	(0.022)
Swedish-speaking										
Cost-share			0.337	(0.263)	0.417	(0.270)	0.588*	(0.269)	0.584*	(0.269)
Lump-sum			−0.078*	(0.035)	−0.085*	(0.035)	−0.105**	(0.036)	−0.109**	(0.036)
Tax credit			0.072*	(0.034)	0.076*	(0.034)	0.090**	(0.034)	0.094**	(0.034)
Foreign-language-speaking										
Cost-share			−1.009	(0.723)	−0.978	(0.737)	−1.119	(0.753)	−1.149	(0.758)
Lump-sum			−0.108*	(0.051)	−0.117*	(0.052)	−0.108*	(0.051)	−0.138**	(0.053)
Tax credit			0.123**	(0.046)	0.130**	(0.046)	0.123**	(0.046)	0.150**	(0.046)
Building age										
Cost-share					−0.025***	(0.005)	−0.026***	(0.005)	−0.026***	(0.005)
Lump-sum					0.000	(0.000)	0.000	(0.000)	0.000	(0.000)
Tax credit					0.000	(0.000)	0.000	(0.000)	0.000	(0.000)
Renovations costs (1000€s)										
Cost-share					0.007	(0.004)	0.007	(0.004)	0.008	(0.004)
Lump-sum					−0.005***	(0.001)	−0.005***	(0.001)	−0.005***	(0.001)
Tax credit					0.005***	(0.001)	0.005***	(0.001)	0.005***	(0.001)
Urban residence area										
Cost-share							0.852***	(0.174)	0.863***	(0.173)
Lump-sum							−0.102***	(0.017)	−0.096***	(0.017)
Tax credit							0.090***	(0.019)	0.085***	(0.019)
Income quintile 2										
Cost-share									−0.260	(0.465)
Lump-sum									−0.124***	(0.029)
Tax credit									0.285***	(0.071)
Income quintile 3										
Cost-share									−0.147	(0.447)
Lump-sum									−0.273***	(0.030)
Tax credit									0.508***	(0.069)
Income quintile 4										
Cost-share									−0.250	(0.451)
Lump-sum									−0.360***	(0.030)
Tax credit									0.612***	(0.069)
Income quintile 5										
Cost-share									−0.114	(0.456)
Lump-sum									−0.576***	(0.033)
Tax credit									0.797***	(0.069)
Observations	14,148		14,148		14,148		14,147		14,147	
McFadden Pseudo-R2	.033		.035		.039		.042		.046	

(*continued on next page*)

Table A.6 (*continued*).

	(1)	(2)	(3)	(4)	(5)
Log likelihood	−10,207	−10,184	−10,143	−10,116	−10,070
AIC	20,433	20,404	20,331	20,280	20,200
BIC	20,509	20,540	20,497	20,461	20,426
Prob > x2	<0.001	<0.001	<0.001	<0.001	<0.001

Robust standard errors in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7
Hausman–McFadden IIA test.

Statistic	Value	p-value
Chi-squared	129.0	0.000

Table A.8
Pairwise probit comparisons of subsidy choice in 2022.
Notes: Binary probit estimates on the subsample of 2022 applicants who chose one of the two instruments in each pair; the dependent variable equals one for the first-listed instrument. Estimates are average semi-elasticities (percentage changes in the predicted probability of selecting the first-listed instrument). Omitted categories: 'Finnish-speaking', 'upper secondary or lower education' and 'non-urban residence areas'.

	(1) Lump-sum vs. tax credit	(2) Cost-share vs. lump-sum	(3) Tax credit vs. cost-share
Disposable income (1000€s)	−0.009***	0.008	0.000
	(0.001)	(0.005)	(0.000)
Reference person age	−0.011***	0.011	0.000
	(0.001)	(0.007)	(0.000)
Single person/parent	0.049*	−0.290	0.001
	(0.021)	(0.240)	(0.005)
Number of children	−0.065***	0.202*	−0.002
	(0.011)	(0.100)	(0.002)
Swedish-speaking	−0.097**	0.645*	−0.014
	(0.035)	(0.279)	(0.009)
Foreign-language-speaking	−0.115*	−1.032	0.017**
	(0.051)	(0.789)	(0.005)
Post-secondary education	−0.066**	0.434	−0.005
	(0.024)	(0.260)	(0.006)
Higher education	−0.080***	0.347	−0.002
	(0.019)	(0.217)	(0.005)
Building age	0.000	−0.029***	0.001***
	(0.000)	(0.006)	(0.000)
Renovations costs (1000€s)	−0.005***	0.067***	0.000
	(0.001)	(0.018)	(0.000)
Urban residence area	−0.092***	0.989***	−0.016***
	(0.017)	(0.181)	(0.004)
Observations	13,987	7483	6824
Pseudo-R2	.041	.082	.043
Log likelihood	−9280	−710	−726
AIC	18,583	1443	1473
BIC	18,666	1519	1548
Prob > x2	<0.001	<0.001	<0.001

Robust standard errors in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.9
Summary statistics.
Notes: "Diff." columns report pairwise differences in means. Standard deviations in parentheses.

	(1) Oil heaters without EPC	(2) Oil heaters with EPC	(3) Subsidy applicants without EPC	(4) Subsidy applicants with EPC	(5) Diff. (1)–(2)	(6) Diff. (3)–(4)
Reference person age	60.15	44.70	56.24	41.52	15.44***	14.72***
	(16.70)	(16.48)	(15.15)	(12.46)	(0.27)	(0.31)
Building age	55.71	50.44	51.04	49.61	5.28***	1.43**
	(19.55)	(17.37)	(20.83)	(18.69)	(0.28)	(0.47)
Urban residence area[a]	0.37	0.49	0.43	0.56	−0.12***	−0.13***
	(0.48)	(0.50)	(0.49)	(0.50)	(0.01)	(0.01)
Higher education[a]	0.33	0.41	0.43	0.56	−0.08***	−0.13***
	(0.47)	(0.49)	(0.50)	(0.50)	(0.01)	(0.01)
Disposable income	35 559.97	36 409.28	41 754.23	39 644.54	−849.31*	2109.70**
	(65 885.18)	(21 299.24)	(55 193.80)	(28 620.73)	(377.45)	(763.86)
Observations	175,092	3855	29,644	1704	178,947	31,348

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
[a] Shares of all households in groups.

Table A.10
Results from baseline probit regression and robustness check.
Notes: (1) Baseline results using building stock registry oil heaters as the control and (2) EPC registry oil heaters as the comparison group. Estimates are average semi-elasticities (percentage changes in the predicted probability of applying).

	(1) Baseline results	(2) Results using EPC data
Disposable income (1000€s)	0.011***	0.007***
	(0.000)	(0.001)
Reference person age	−0.007***	0.001
	(0.001)	(0.002)
Single person/parent	−0.414***	−0.295***
	(0.015)	(0.055)
Number of children	−0.015*	0.132***
	(0.007)	(0.020)
Swedish-speaking	−0.178***	−0.064
	(0.023)	(0.100)
Foreign-language-speaking	−0.178***	−0.205**
	(0.034)	(0.076)
Post-secondary education	0.186***	0.039
	(0.017)	(0.080)
Higher education	0.171***	0.213***
	(0.014)	(0.043)
Building age	−0.009***	−0.002*
	(0.000)	(0.001)
Years lived in house	−0.016***	−0.020***
	(0.001)	(0.003)
House rented	−0.069*	−2.241**
	(0.035)	(0.758)
Peri-urban residence area	−0.088***	−0.114*
	(0.014)	(0.045)
Rural residence area	−0.146***	−0.174**
	(0.014)	(0.055)
Observations	600,348	4585
Pseudo-R2	.04	.08
Log likelihood	−118,098	−2783
AIC	236,226	5597
BIC	236,396	5693
Prob > x2	<0.001	<0.001
Year fixed-effects	Yes	Yes

Standard errors clustered at household level in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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